JobAI: EDA on the ML-ready fraud dataset (v2)

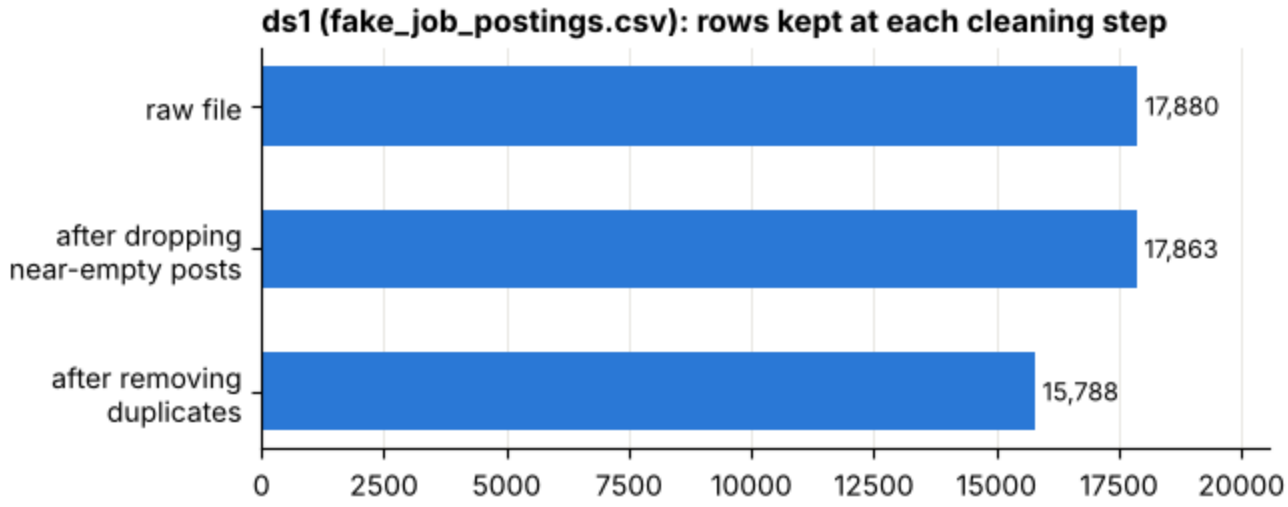
Checkpoint 2. Target: `fraudulent` (1 = fake posting). Dataset built by `src/data/build_dataset.py`.

20,288 postings x 22 columns

	dtype	missing_%	unique	example
posting_id	str	0.0	20288	p000001
source	str	0.0	3	ds1_labeled
label_source	str	0.0	2	labeled
fraudulent	int64	0.0	2	1
title	str	0.0	14564	Cruise Staff Wanted *URGENT*
description	str	0.0	19648	6* Ultra Luxury American Cruise Company
employment_type	str	0.0	7	unknown
has_company	int64	0.0	2	0
has_location	int64	0.0	2	1
has_salary	int64	0.0	2	0
has_experience	int64	0.0	2	0
has_employment_type	int64	0.0	2	0
desc_words	int64	0.0	1237	256
title_words	int64	0.0	22	4
has_url	int64	0.0	2	0
has_email	int64	0.0	2	1
has_phone	int64	0.0	2	0
money_words	int64	0.0	2	1
title_all_caps	int64	0.0	2	0
company_key	str	0.0	7835	nocompany_p000001
split	str	0.0	3	train
label	str	0.0	2	fraud

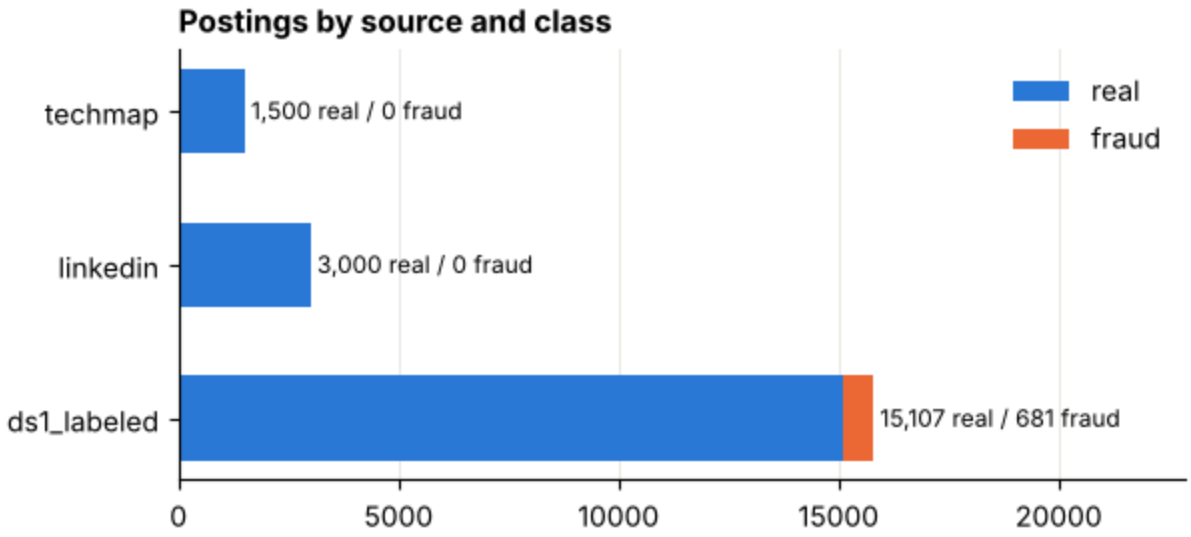
1. Lineage: how the dataset was built

	raw	short_dropped	dupes_dropped	kept	used_in_v2
ds1_labeled	17880	17	2075	15788	15788
linkedin	20000	14	914	19072	3000
techmap	33064	24	6678	26362	1500



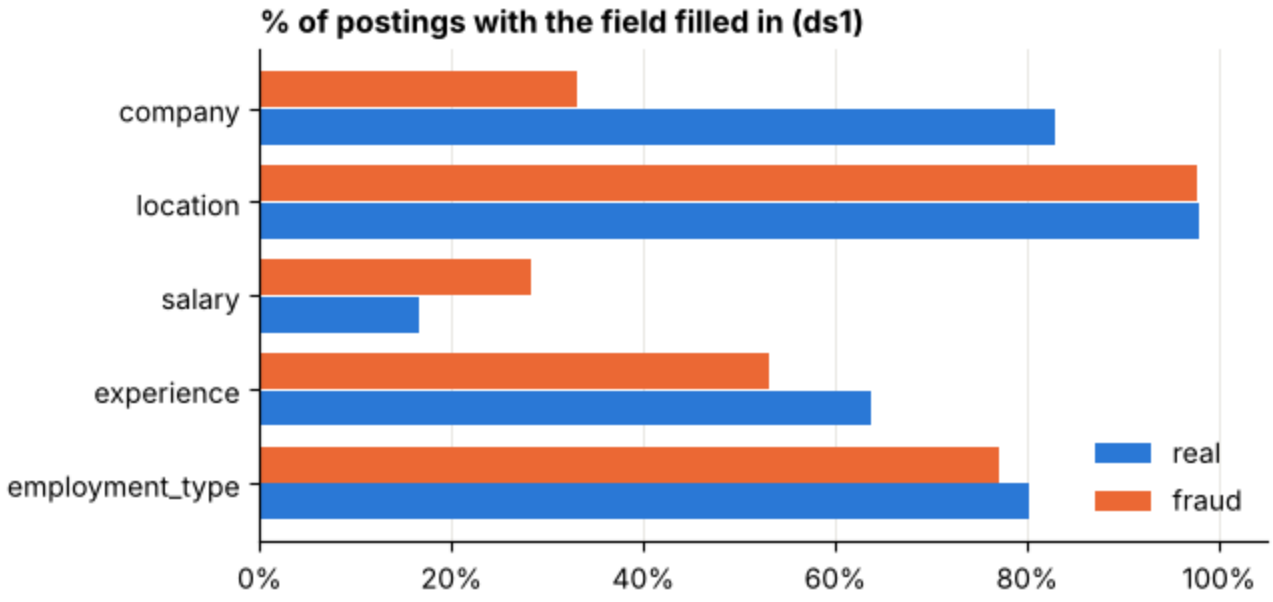
2. Class balance and sources

split	rows	fraud	fraud_%
train	14492	487	3.36
val	2898	97	3.35
test	2898	97	3.35



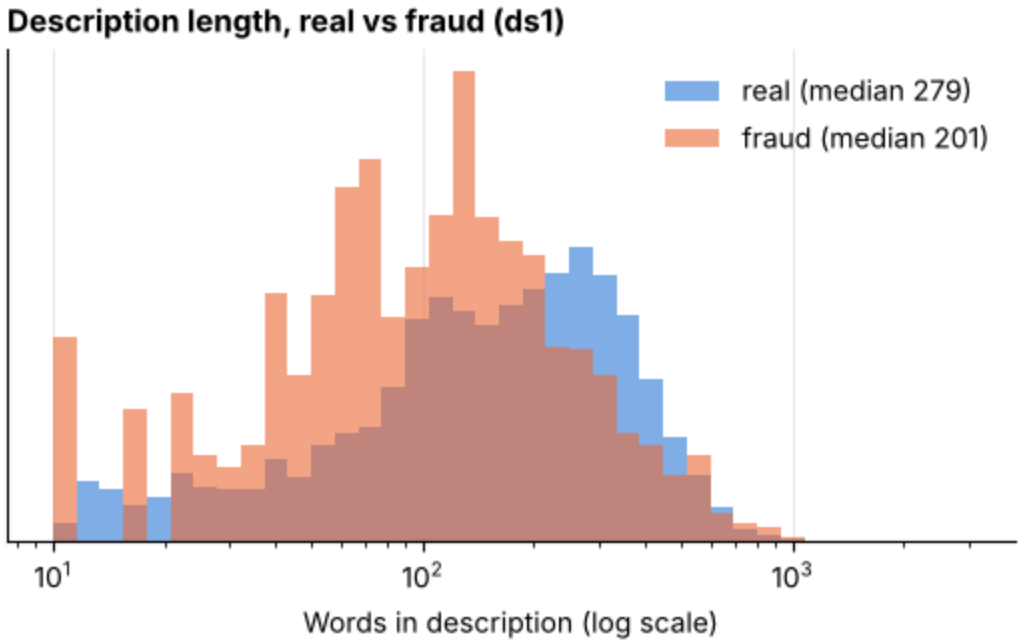

```
overall fraud rate: 3.4%
```

3. Missing fields by class (feature vs target)



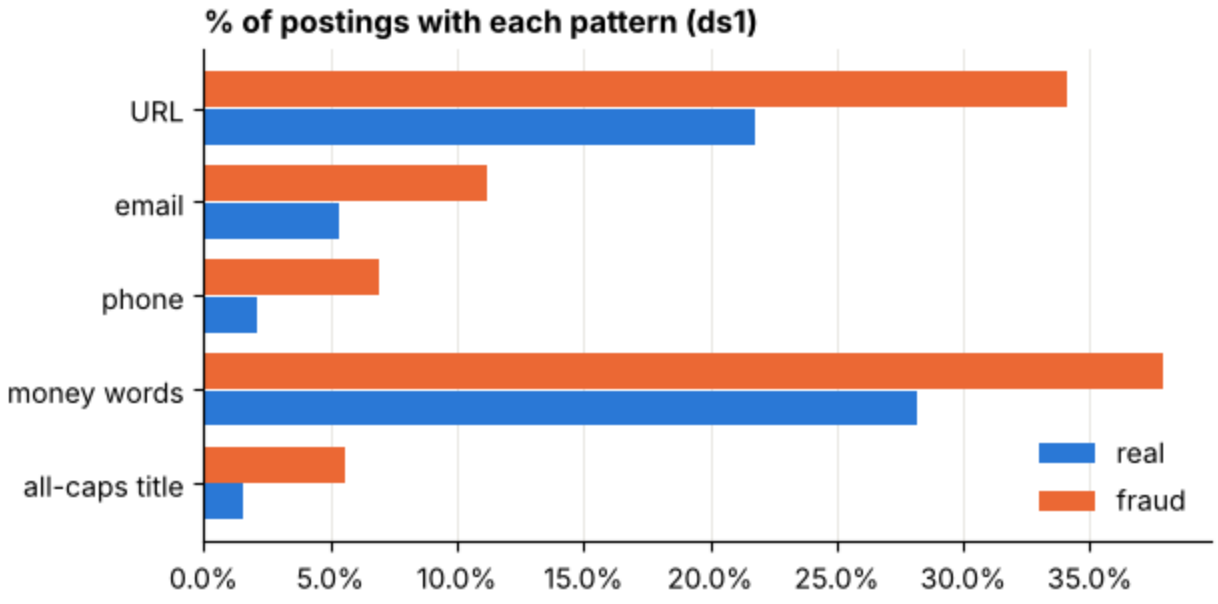
label	real	fraud
has_company	82.9	33.0
has_location	97.9	97.8
has_salary	16.5	28.2
has_experience	63.7	53.2
has_employment_type	80.1	77.1

4. Description length by class



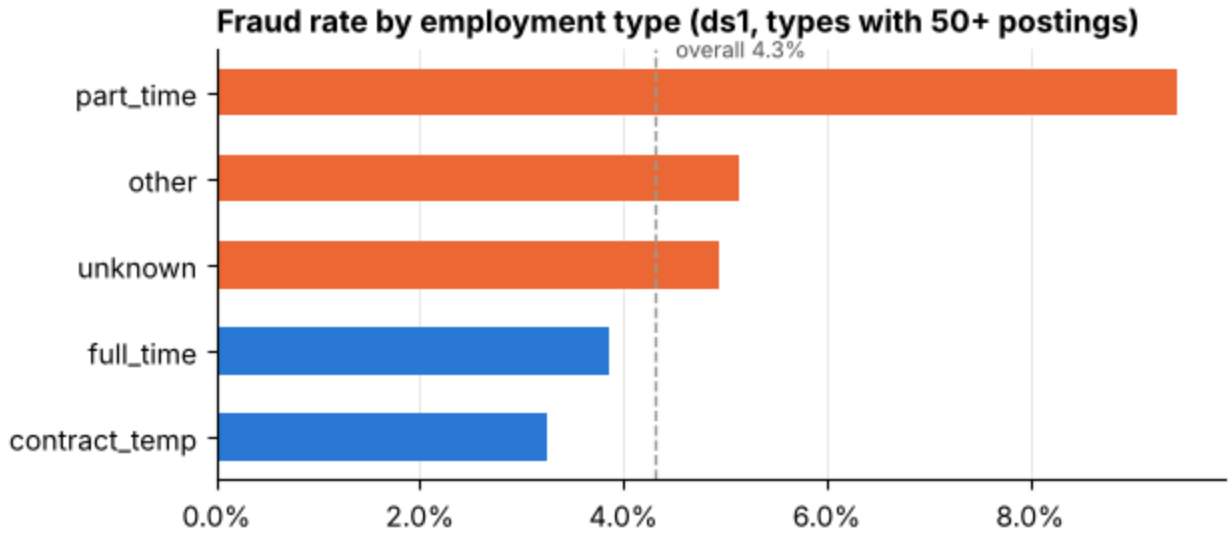
label	count	mean	std	min	25%	50%	75%	max
fraud	681.0	264.0	196.0	10.0	122.0	201.0	359.0	1347.0
real	15107.0	302.0	169.0	10.0	178.0	279.0	393.0	2115.0

5. Text patterns by class



label	real	fraud	fraud / real
has_url	21.8	34.1	1.6
has_email	5.3	11.2	2.1
has_phone	2.1	6.9	3.3
money_words	28.1	37.9	1.3
title_all_caps	1.5	5.6	3.6

6. Employment type by class

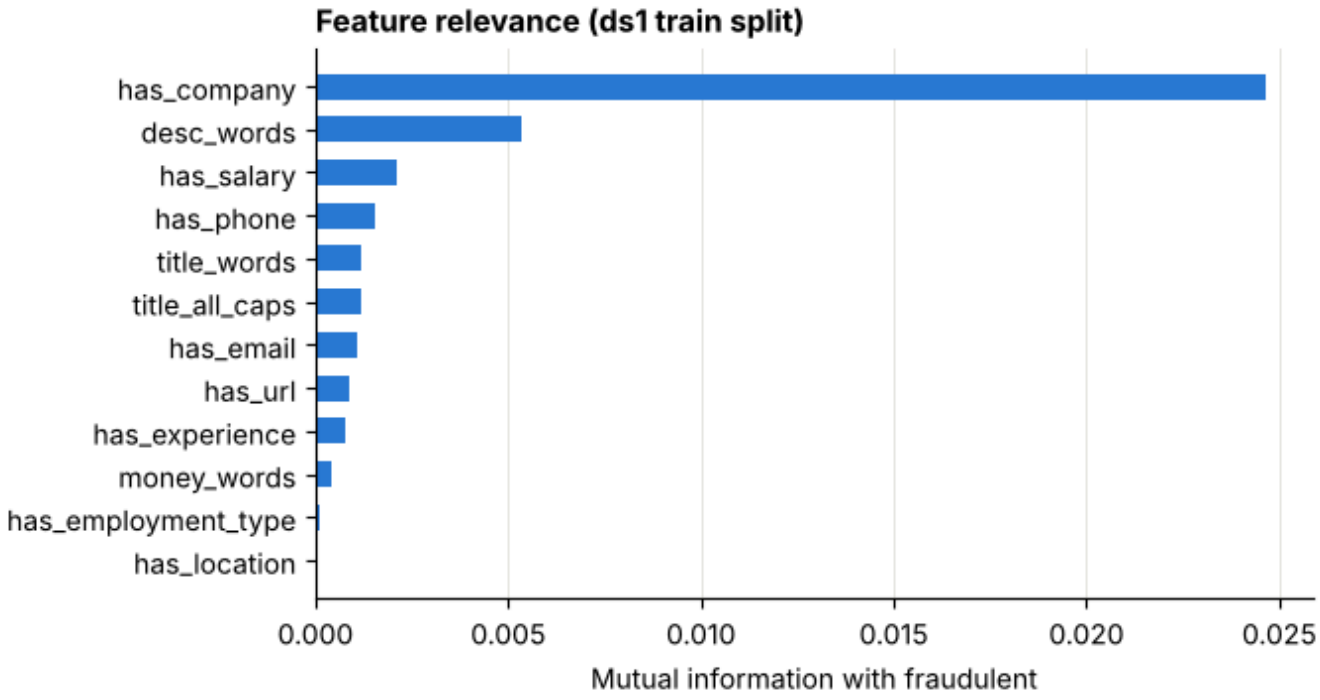


employment_type	postings	fraud_rate
contract_temp	1049	3.2
full_time	10626	3.9
unknown	3157	4.9
other	214	5.1
part_time	742	9.4

7. Words that separate real and fraud

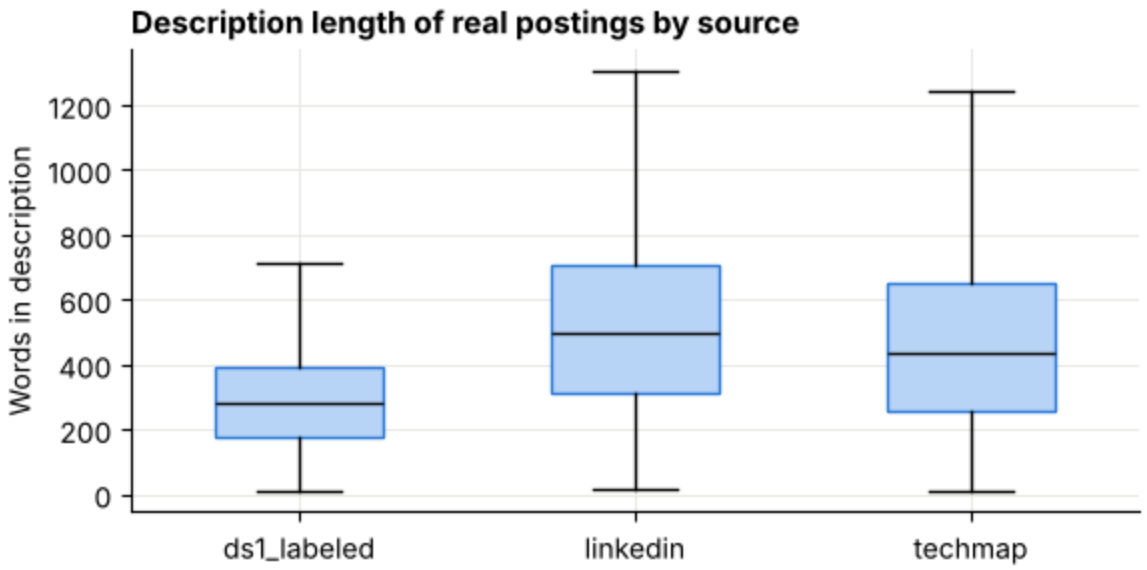
	more common in fraud	x	more common in real	x
0	lawson	241.6	amazing	29.1
1	darren	120.8	excited	20.8
2	distractions	69.5	touch	18.6
3	hse	51.2	gym	17.1
4	odm	47.1	keen	15.9
5	wages	24.0	custom	15.6
6	dsl	24.0	investors	15.2
7	faces	23.4	nice	14.8
8	motors	22.0	european	13.2
9	plc	20.6	stuff	13.0
10	clerk	16.8	native	12.8
11	diem	15.5	remuneration	12.4
12	afternoon	14.1	awesome	12.3
13	preventive	13.2	seek	12.0
14	bodies	12.5	php	11.4

8. Which features relate most to the target



	mutual_info	correlation
has_company	0.0246	-0.256
desc_words	0.0054	-0.056
has_salary	0.0021	0.070
has_phone	0.0015	0.067
title_words	0.0012	-0.008
title_all_caps	0.0012	0.059
has_email	0.0011	0.052
has_url	0.0009	0.043
has_experience	0.0008	-0.040
money_words	0.0004	0.028
has_employment_type	0.0001	-0.013
has_location	0.0001	0.011

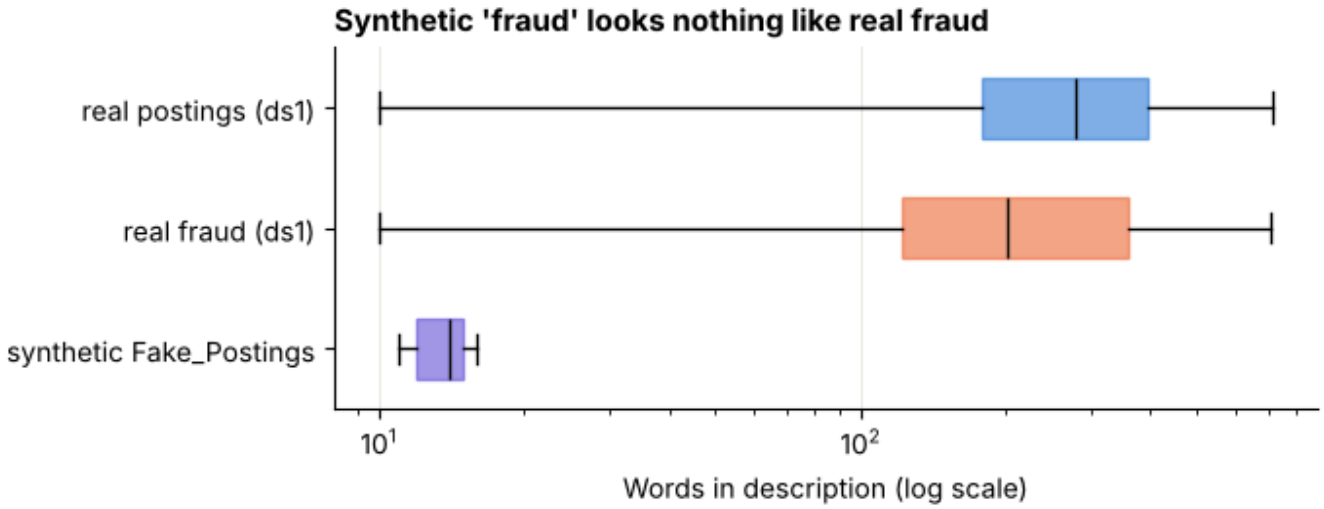
9. Source check: can a model tell the sources apart?



source	desc_words	has_company	has_salary	has_email	money_words
ds1_labeled	301.77	0.83	0.17	0.05	0.28
linkedin	532.29	0.98	0.31	0.12	0.56
techmap	484.17	1.00	0.29	0.06	0.60

10. Why the synthetic Fake_Postings data was removed

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synthetic rows: 10,000; share with the same 'Earn $5000/week! Immediate hiring' line:
100%
synthetic description length: median 14 words (min 11, max 16)
real fraud description length (ds1): median 201 words
```



	title	description	company_profile	location
0	Mental health nurse	Arm drive court sure vote. Earn $5000/week! Im...	Rivera and Sons - Established 2022.	West Jeffrey
1	Conference centre manager	Government whom its bed go tax tree black. Ear...	Davidson, Jones and Gomez - Established 2003.	Lake Meredithberg
2	Engineer, land	I member discuss follow way there nation. Earn...	Allen Ltd - Established 1998.	Lake Cathybury

	model	precision	recall	f1	pr_auc
2	majority	0.000	0.000	0.000	0.042
5	rule	0.121	0.639	0.204	0.093
8	tfidf_lr	0.609	0.866	0.715	0.834
11	tfidf_lr_tuned_threshold	0.866	0.732	0.793	0.834
14	tfidf_lr_with_synthetic	0.889	0.495	0.636	0.812

11. Key findings

1. Fraud is rare: 3.4% of postings overall (4.3% within ds1). Accuracy is misleading, so we use precision, recall, F1 and PR-AUC.
2. Missing company info is the strongest single signal: 83% of real ds1 postings have a company profile vs 33% of fraud.

3. Fraud postings are shorter (median 201 vs 279 words) and more often contain phone numbers (3.3x), all-caps titles (3.6x), emails (2.1x) and URLs. Fraud lists a salary more often (28% vs 17%), and part-time jobs have the highest fraud rate (9.4%).
4. Many of the strongest fraud words are specific names (likely recruiters or companies that posted many scams), so a model can memorize them instead of learning general scam patterns. We split by company and will check performance on unseen companies.
5. Real postings from LinkedIn and Techmap are longer than ds1's, so we always report results on the ds1-only test set to avoid rewarding source detection.
6. The synthetic Fake_Postings file is one repeated template (about 14 words). Training with it cut recall on real scams from 87% to 50%, so it was removed.

Dataset card

Dataset card: JobAI fraud postings v2 (draft)

Summary

Job postings labeled real (0) or fraudulent (1), built to train the JobAI fraud filter. 20,288 postings, 681 fraudulent (3.4%). Split 70/15/15 into train/val/test.

Sources

Source	Rows used	Label	Notes
ds1: Kaggle fake_job_postings.csv (17,880 rows, 866 fraud)	15,107 real, 681 fraud after cleaning	labeled	Only labeled fraud source. 2,075 duplicates removed
LinkedIn Job Postings 2023-24 (arshkon, Kaggle)	3,000	assumed real	Random sample from a 20k pool
US Job Postings 2023-05-05 (techmap, Kaggle)	1,500	assumed real	Random sample, flattened from nested JSON
Synthetic Fake_Postings.csv (10,000 rows)	0 (excluded)	labeled fraud	Generated text with one repeated template; only used in a comparison run

Lineage (src/data/build_dataset.py)

1. Load each source (ds1 description = description + requirements + benefits) and map it to one schema: title, description, company, location, employment type, experience, salary.
2. Clean text: unescape HTML, strip tags, split glued words, replace URLs, emails and phone numbers with `urltoken`, `emailtoken`, `phonetoken` in every source.
3. Drop descriptions under 10 words; drop exact duplicate title + description.
4. Add 3,000 LinkedIn and 1,500 Techmap real postings (random sample, seed 42).
5. Engineer features (below).
6. Split 70/15/15 with StratifiedGroupKFold: fraud share equal in each split, no company appears in more than one split. Seed 42.

Fields

Field	Type	Meaning
posting_id	str	Row id
source	str	ds1_labeled, linkedin, techmap
label_source	str	labeled or assumed_real

Field	Type	Meaning
fraudulent	int	Target: 1 = fraud
title, description	str	Cleaned text
employment_type	str	full_time, part_time, contract_temp, internship, full_or_part_time, other, unknown
has_company, has_location, has_salary, has_experience, has_employment_type	0/1	Field is filled in
has_url, has_email, has_phone	0/1	Description contains one
money_words	0/1	Mentions $ amounts, "earn", "per week", "cash"
title_all_caps	0/1	Title written in all caps
desc_words, title_words	int	Word counts
company_key	str	Hashed company, used for grouped splitting

Known issues and limitations

- Small fraud set: 681 examples, 97 in test, so metrics have wide error bars.
- LinkedIn and Techmap labels are assumed real; a few scams may be in there.
- Source differences: ds1 descriptions are shorter than LinkedIn/Techmap, and ds1 "company" is a company profile, not a name. Report metrics on ds1-only test to avoid rewarding source detection.
- Fraud examples come from one older dataset, so new scam styles may not be covered.
- English, mostly U.S. postings.

Intended use

Training and evaluating the JobAI fraud warning. Not for judging specific employers.

License

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